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# =====
# 3-STOREY MASONRY FRAME
# GRAVITY + FOUNDATION SETTLEMENT + PUSHOVER
# =====

wipe
model basic -ndm 3 -ndf 6

# =====
# GEOMETRY
# =====

set H_story 3.0
set L_span 3.0

set L_pier 1.0
set T_pier 0.25
set H_span 0.80

# =====
# MATERIAL
# =====

set E 1700.0e6
set G 550.0e6
set fc 6.0e6
set c 0.150e6
set Gc 6.0
set mu0 0.40
set beta 0.30

set rho 1200.0
set g 9.81

# =====
# NODES
# =====

node 1 0.0 0.0 0.0
node 2 $L_span 0.0 0.0

node 3 0.0 0.0 $H_story
node 4 $L_span 0.0 $H_story

node 5 0.0 0.0 [expr 2.0*$H_story]
node 6 $L_span 0.0 [expr 2.0*$H_story]

node 7 0.0 0.0 [expr 3.0*$H_story]
node 8 $L_span 0.0 [expr 3.0*$H_story]

node 9 0.0 0.0 [expr 0.5*$H_story]
node 10 $L_span 0.0 [expr 0.5*$H_story]

node 11 0.0 0.0 [expr 1.5*$H_story]
node 12 $L_span 0.0 [expr 1.5*$H_story]

node 13 0.0 0.0 [expr 2.5*$H_story]
node 14 $L_span 0.0 [expr 2.5*$H_story]

node 15 [expr $L_span/2.0] 0.0 $H_story
node 16 [expr $L_span/2.0] 0.0 [expr 2.0*$H_story]
node 17 [expr $L_span/2.0] 0.0 [expr 3.0*$H_story]

# =====
# BOUNDARY CONDITIONS
# =====

fix 1 1 1 1 1 1
fix 2 1 1 1 1 1

# =====
# INTERNAL NODE CONSTRAINTS

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# Removes zero-stiffness DOFs that cause a singular stiffness matrix.
# Pier nodes (9-14): element axis = Z → fix Y, Rx, Rz
# Spandrel nodes (15-17): element axis = X → fix Y, Z, Rx, Rz
# DOF order: X Y Z Rx Ry Rz
# =====

foreach n {9 10 11 12 13 14} {
    fix $n 0 1 0 1 0 1
}

foreach n {15 16 17} {
    fix $n 0 1 1 1 0 1
}

# =====
# PIER ELEMENTS
# =====

element Macroelement3d 1 \
    1 3 9 \
    0.0 0.0 1.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_story $L_pier $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

element Macroelement3d 2 \
    2 4 10 \
    0.0 0.0 1.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_story $L_pier $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

element Macroelement3d 3 \
    3 5 11 \
    0.0 0.0 1.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_story $L_pier $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

element Macroelement3d 4 \
    4 6 12 \
    0.0 0.0 1.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_story $L_pier $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

element Macroelement3d 5 \
    5 7 13 \
    0.0 0.0 1.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_story $L_pier $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

```

```

element Macroelement3d 6 \
    6 8 14 \
    0.0 0.0 1.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_story $L_pier $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

# =====
# SPANDREL ELEMENTS
# =====

element Macroelement3d 7 \
    3 4 15 \
    1.0 0.0 0.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_span $L_span $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

element Macroelement3d 8 \
    5 6 16 \
    1.0 0.0 0.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_span $L_span $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

element Macroelement3d 9 \
    7 8 17 \
    1.0 0.0 0.0 \
    0.0 1.0 0.0 \
    -tremuri \
    $H_span $L_span $T_pier \
    $E $G $fc $mu0 $c $Gc $beta \
    -density $rho \
    -cmass \
    -pDelta

# =====
# RECORDERS
# =====

recorder Node      -file RoofDisp.out      -time -node 8      -dof 1 disp
recorder Node      -file SupportReaction.out -time -node 1 2    -dof 1 reaction
recorder Element   -file PierForces.out     -time -ele 1 2 3 4 5 6 basicForce
recorder Element   -file SpanForces.out     -time -ele 7 8 9    basicForce

# =====
# GRAVITY LOADS
#
# IMPORTANT – choose ONE of the two floor load definitions below
# and comment out the other, depending on what the load represents:
#
# Option A: floor load as a surface pressure of 5 kN/m²
#   set floorLoad [expr -5.0e3 * $L_span * $T_pier]      ;# -3 750 N per node
#
# Option B: floor load as weight of 5 m of masonry wall (tributary height = 5 m)
#   set floorLoad [expr -5.0*$g*$rho*$L_span*$T_pier]    ;# -44 145 N per node
# =====

set floorLoad [expr -5.0*$g*$rho*$L_span*$T_pier]

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```

pattern Plain 10 Linear {
    for {set e 1} {$e <= 9} {incr e} {
        eleLoad -ele $e -type -selfWeight 0.0 0.0 [expr -$g]
    }

    load 3 0.0 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 4 0.0 0.0 0.0 $floorLoad 0.0 0.0 0.0

    load 5 0.0 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 6 0.0 0.0 0.0 $floorLoad 0.0 0.0 0.0

    load 7 0.0 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 8 0.0 0.0 0.0 $floorLoad 0.0 0.0 0.0
}

system BandGeneral
numberer Plain
constraints Transformation

test NormUnbalance 1.0 50 0
algorithm Newton

integrator LoadControl 0.01
analysis Static

puts "Applying gravity loads..."

analyze 100

loadConst -time 0.0

# =====
# FOUNDATION SETTLEMENT
# =====

pattern Plain 20 Linear {
    sp 1 3 -0.001
}

integrator LoadControl 0.01

puts "Applying settlement..."

analyze 200

loadConst -time 0.0

record

# =====
# PUSHOVER LOAD PATTERN
# =====

pattern Plain 30 Linear {

    load 3 0.20 0.0 0.0 0.0 0.0 0.0
    load 4 0.20 0.0 0.0 0.0 0.0 0.0

    load 5 0.40 0.0 0.0 0.0 0.0 0.0
    load 6 0.40 0.0 0.0 0.0 0.0 0.0

    load 7 0.60 0.0 0.0 0.0 0.0 0.0
    load 8 0.60 0.0 0.0 0.0 0.0 0.0
}

# =====
# PUSHOVER ANALYSIS
# =====

constraints Transformation

```

```
numberer Plain
system BandGeneral

test NormDispIncr 1.0e-6 100 0
algorithm Newton

analysis Static

set targetDisp 0.20
set incr      0.0001

set nSteps [expr int($targetDisp/$incr)]

integrator DisplacementControl 8 1 $incr

puts "Running pushover..."

set completedSteps 0
set usingInitial 0

while {$completedSteps < $nSteps} {

    set ok [analyze 1]

    if {$ok == 0} {
        incr completedSteps
        if {$usingInitial} {
            algorithm Newton
            test NormDispIncr 1.0e-6 100 0
            set usingInitial 0
        }
    } else {
        if {!$usingInitial} {
            puts "Step $completedSteps: Newton failed – switching to Newton -initial"
            test NormDispIncr 1.0e-5 200 5
            algorithm Newton -initial
            set usingInitial 1
        }
        set ok [analyze 1]
        if {$ok == 0} {
            incr completedSteps
        } else {
            puts "Analysis stopped at step $completedSteps / $nSteps"
            puts "Roof displacement at stop: [expr $completedSteps * $incr] m"
            break
        }
    }
}

if {[expr $completedSteps % 200] == 0 && $completedSteps > 0} {
    puts " Step $completedSteps / $nSteps ([expr $completedSteps * $incr] m)"
}

puts "Pushover completed: $completedSteps of $nSteps steps."
puts "Final roof displacement: [expr $completedSteps * $incr] m"
puts "Results: RoofDisp.out SupportReaction.out PierForces.out SpanForces.out"
```